

$$\{A\} = ([Z]^T[Z])^{-1}[Z]^T\{Y\}$$

This equation gives the coefficients of a linear least-squares fit.
Consider the linear model:

$$y = a_0 + a_1x$$

Given the data:

$$\begin{array}{c|c} x & y \\ \hline 1 & 2 \\ 2 & 3 \\ 3 & 5 \end{array}$$

The coefficient vector is

$$\{A\} = \begin{bmatrix} a_0 \\ a_1 \end{bmatrix}$$

The observation vector is

$$\{Y\} = \begin{bmatrix} 2 \\ 3 \\ 5 \end{bmatrix}$$

The design matrix is

$$[Z] = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$$

The transpose of the design matrix is

$$[Z]^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$$

The normal matrix is

$$[Z]^T[Z] = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

The inverse of the normal matrix is

$$([Z]^T[Z])^{-1} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix}$$

Finally, the coefficient vector is

$$\{A\} = \begin{bmatrix} 0.33 \\ 1.50 \end{bmatrix}$$

Hence, the least-squares regression line is

$$y = 0.33 + 1.50x$$

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We want the root of

$$x^2 e^{-x/2} = 1$$

in the interval $[0, 2]$. Define

$$f(x) = x^2 e^{-x/2} - 1.$$

The secant iteration formula is

$$x_{n+1} = x_n - \frac{f(x_n)(x_n - x_{n-1})}{f(x_n) - f(x_{n-1})}.$$

Taking $x_0 = 0$, $x_1 = 2$, the iterations are:

Iter	x_{n-1}	x_n	x_{n+1}	$f(x_{n+1})$
1	0.0000	2.0000	1.3591	-0.0637
2	2.0000	1.3591	1.4355	0.0053
3	1.3591	1.4355	1.4296	0.00003
4	1.4355	1.4296	1.4296	-0.000000018

Thus, the method converges to

$$x \approx 1.430 \quad (\text{correct to three decimal places}).$$

2 Problem

The velocity y of a falling parachutist is given by

$$y(t) = \frac{g m}{c} \left(1 - e^{-\frac{c}{m}t} \right),$$

where $g = 9.81 \text{ m/s}^2$.

For a parachutist with a drag coefficient $c = 15 \text{ kg/s}$, compute the mass m so that the velocity is

$$y = 36 \text{ m/s} \quad \text{at} \quad t = 10 \text{ s.}$$

Use the **false-position method** to determine m to a level of $\varepsilon_s = 0.1\%$.

2.1 Solution

Define the root-finding function

$$f(m) = \frac{g m}{c} (1 - e^{-\frac{c}{m} t}) - y_{\text{target}}.$$